STQA-5: Selenium Grid and WebDriver with Java in Eclipse

Installation checklist

Step	Action	Check
1	Install JDK and run `java -version`	Version is printed
2	Install Eclipse IDE, choose a workspace folder	Eclipse opens
3	Download Selenium Java client JAR and its libs; download ChromeDriver matching your Chrome version	Files in a known folder
4	File > New > Java Project; create a package and a class	Project shows in Package Explorer
5	Project > Properties > Java Build Path > Add External JARs: add the client JAR and all JARs in libs	No red errors on import lines
6	Run the test script below	Browser opens and title is printed

Verification script (FirstTest.java)

```
package demo;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class FirstTest {
    public static void main(String[] args) {
        System.setProperty("webdriver.chrome.driver", "C:\\drivers\\chromedriver.exe");
        WebDriver driver = new ChromeDriver();
        driver.get("https://www.google.com");
        System.out.println("Page title: " + driver.getTitle());
        driver.quit();
    }
}
```

Expected output in the Eclipse console

```
Page title: Google
```

Selenium Grid (hub and node) commands

```
# Selenium 3.x Grid
java -jar selenium-server-standalone-3.141.59.jar -role hub
java -jar selenium-server-standalone-3.141.59.jar -role node -hub http://localhost:4444/grid/register
# open http://localhost:4444/grid/console to see the registered node
```